

EventShield Pro

Biometric Access Control System

Complete System Requirements Document

Organization: The Super 6 Tournament Management

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Classification: Confidential — Internal Use Only

1. Executive Summary

EventShield Pro is a venue access control system that integrates QR code ticketing with biometric facial recognition to manage entry into The Super 6 tournament events. The system eliminates manual ticket checking, reduces entry bottlenecks, and prevents ticket fraud by binding each ticket to the purchaser's face.

The system operates in three stages: (1) ticket purchase and QR code delivery, (2) biometric face registration at entry, and (3) automated gate access via facial recognition at the turnstile. All biometric data is stored on-device only, with no cloud transmission.

2. System Overview

2.1 Ticket Types

ID	Requirement	Priority	Notes
TK-01	Daily Ticket — grants access for one calendar day only	MUST	Access window: event day 6am–midnight
TK-02	Weekend Ticket — grants access for Saturday and Sunday	MUST	Both days included, no additional purchase
TK-03	Staff Ticket — grants unlimited access with no expiry	MUST	Issued to workers, volunteers, officials

2.2 Purchase Channels

ID	Requirement	Priority	Notes
PC-01	Web portal ticket purchase with QR code emailed/texted to customer	MUST	Must support mobile browsers
PC-02	On-site POS terminal ticket purchase with printed QR code	MUST	Receipt printer required at counter
PC-03	QR code must encode ticket type, unique ID, and issue timestamp	MUST	Required for scanner validation
PC-04	QR codes must be non-transferable and single-use for registration	MUST	Prevents ticket sharing
PC-05	Once QR code is scanned and access is granted, the QR code must display 'USED' — large, bold, and diagonally across the QR code image in red. This must trigger immediately upon successful gate entry and persist on the ticket permanently.	MUST	Prevents reuse — visual invalidation required on phone + printout

2.3 QR Code Delivery

ID	Requirement	Priority	Notes
QR-01	QR code delivered via SMS to customer phone number	MUST	Immediate upon purchase
QR-02	QR code delivered via email as image attachment	MUST	Backup delivery method
QR-03	QR code printable from web portal	SHOULD	For customers without smartphones
QR-04	POS terminal prints QR code on thermal receipt paper	MUST	Required for counter sales

3. Biometric Registration — Step 2

3.1 Hardware Requirements

ID	Requirement	Priority	Notes
HW-01	Facial recognition kiosk with integrated QR code scanner	MUST	Floor-standing or wall-mount
HW-02	Minimum camera resolution: 1080p with IR night vision	MUST	For low-light venues
HW-03	QR scanner must support Code 128, QR Code, Data Matrix formats	MUST	Standard barcode compatibility
HW-04	Device must have local onboard storage (min. 256GB SSD)	MUST	No cloud dependency

ID	Requirement	Priority	Notes
HW-05	Device must operate offline with no internet required	MUST	Venues may lack reliable WiFi
HW-06	Touchscreen display minimum 10 inches for face alignment UI	SHOULD	Improves capture accuracy

3.2 Registration Flow

ID	Requirement	Priority	Notes
RF-01	Customer presents QR code to scanner — system reads ticket type	MUST	First action at kiosk
RF-02	System displays face capture screen with circular alignment guide	MUST	Visual UI must be clear and intuitive
RF-03	System provides audio/visual cue when face is properly aligned	SHOULD	Reduces failed captures
RF-04	System captures and stores face biometric data within 3 seconds	MUST	Max acceptable capture time
RF-05	Biometric data linked to ticket ID and ticket type in local DB	MUST	Required for gate validation
RF-06	If face capture fails 3 times, staff alert is triggered	MUST	Fallback for registration failures
RF-07	First-time customers only — returning customers skip to gate	MUST	If face already registered for ticket

3.3 Access Rules by Ticket Type

ID	Requirement	Priority	Notes
AR-01	Daily ticket face registration grants access only on date of purchase	MUST	System checks date at every scan
AR-02	Weekend ticket face registration grants access Sat AND Sun	MUST	Both days loaded at registration
AR-03	Staff ticket face registration grants unlimited access — no date check	MUST	Exempt from date validation
AR-04	Expired tickets must be rejected at gate — no exceptions	MUST	Daily tickets expire at midnight

4. Gate Access — Step 3

4.1 Turnstile Hardware

ID	Requirement	Priority	Notes
TS-01	Tripod or full-height turnstile with electronic lock mechanism	MUST	Standard venue turnstile
TS-02	Turnstile connected to facial recognition scanner via wired/wireless signal	MUST	Gate opens on scanner approval
TS-03	Turnstile must default to LOCKED state on power cycle	MUST	Fail-secure design
TS-04	Manual override key available for staff emergency use	MUST	Required by fire code
TS-05	Gate open duration: 3–5 seconds per entry	MUST	Prevents tailgating

4.2 Gate Scan Flow

ID	Requirement	Priority	Notes
GS-01	Customer approaches turnstile — face scanner activates automatically	MUST	Motion or proximity triggered
GS-02	Scanner matches face against local biometric database within 1 second	MUST	Max acceptable match time
GS-03	Successful match sends unlock signal to turnstile	MUST	Core gate function
GS-04	Green light and audio chime on successful entry	MUST	Clear feedback to customer
GS-05	Failed match displays red light — gate remains locked	MUST	With error message on screen
GS-06	Failed match after 3 attempts sends alert to staff device	MUST	Staff notified via app or buzzer
GS-07	System logs every entry attempt with timestamp and ticket ID	MUST	For audit trail

5. Data Storage & Privacy

ID	Requirement	Priority	Notes
DP-01	All biometric data stored on local device only — no cloud upload	MUST	Privacy compliance requirement
DP-02	Biometric data encrypted at rest using AES-256	MUST	Security requirement
DP-03	Data automatically purged 30 days after event end	MUST	Data minimization principle
DP-04	No PII (names, emails) stored with biometric data — ticket ID only	MUST	Anonymization requirement

ID	Requirement	Priority	Notes
DP-05	Staff access to biometric data requires admin password	MUST	Role-based access control
DP-06	Customers must be informed of face capture via posted signage	MUST	Legal notice requirement

6. Admin Dashboard Requirements

ID	Requirement	Priority	Notes
AD-01	Real-time entry count per event displayed on dashboard	MUST	Capacity management
AD-02	Ability to revoke any ticket type immediately from dashboard	MUST	For fraud or bans
AD-03	Export entry logs as CSV for post-event reporting	SHOULD	Finance and compliance
AD-04	Failed entry alerts shown on admin dashboard in real time	MUST	Security monitoring
AD-05	Dashboard accessible from tablet or laptop on local network	MUST	Venue staff use

7. Performance Requirements

ID	Requirement	Priority	Notes
PF-01	System must process minimum 200 entries per hour per gate	MUST	For peak entry periods
PF-02	Face match accuracy must exceed 99.5% for registered faces	MUST	Industry standard
PF-03	False acceptance rate (FAR) must be below 0.01%	MUST	Security threshold
PF-04	System must function in temperatures 40°F–105°F	MUST	Florida outdoor venues
PF-05	Device battery backup minimum 8 hours for portable deployment	SHOULD	For outdoor venues without power

8. Priority Key

MUST

Non-negotiable. System cannot function correctly without this requirement.

SHOULD	Strongly recommended. Should be included unless constrained by time or budget.
COULD	Nice to have. Include if resources permit. Will not impact core functionality.

End of Document — EventShield Pro System Requirements v1.0